

Comparing Long-term Returns from Constant Mix and Buy-Hold Strategies

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1. INTRODUCTION

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1.1 INTRODUCTION: BACKGROUND (1/2)

In 2021, the Federal Reserve Board reports that three-fourths of non-retired adults had at least some retirement savings in the US.

Among those with retirement savings, these savings were most frequently in defined contribution plans, such as a 401(k) or 403(b), with 55% of non-retired adults having money in such a plan.

Nowadays, 401k contribution plans allow individuals to actively participate in customizing and electing their own investment strategies and asset allocations.

1.1 INTRODUCTION: BACKGROUND (2/2)

Understanding retirement/long-term investing and how to invest is beneficial in building wealth and financial freedom over time.

It can be quite challenging to decide which strategies and assets to elect, especially since the stock market is characteristically volatile and dependent on many factors.

In addition, there are many investment strategies and assets choices, all with various risks and returns.

One type of investment strategy is rebalancing strategy.

Rebalancing strategy reduces risk and enhances returns through periodically buying and selling assets to maintain the desired allocation. Common simple rebalancing strategies include buy-hold and constant-mix.

PURPOSE OF STUDY

This study aims to compare the performance of the buy-hold strategy and the 40% bond/60% stock rebalancing strategy using historical simulation from August 2002 to August 2022, a 20 year period.

I will also compare the returns of different asset allocations, which includes small cap (IWM), technology (QQQ), large cap (SPY), and bonds (TLT).

For each strategy, contributions will be made monthly.

For the buy-hold strategy, stocks accumulated each month will never be sold until the end of August 2022.

For the 40/60 constant mix strategy, rebalancing will take place at the end of each month until end of August 2022.

1.2 INTRODUCTION: BUY-HOLD STRATEGY (1/2)

Traditionally, investors would try to maximize their gains and minimize their losses through buying stock when the stock price is low and selling stocks when the stock price is high.

However, this “buy low and sell high” strategy requires intuition, knowledge, and timing to execute successfully.

The Buy-hold strategy is a passive, long-term investment strategy where one buys stocks and holds on to the stock for a long periods of time. Buy-hold strategy allows the individual to invest successfully with little management.

1.2 INTRODUCTION: BUY-HOLD STRATEGY (2/2)

The strategy's success lies in the long-term growth of stocks over the years despite market fluctuations.

So far, the market has been able to recover from declines and overall has positive annual returns.

One disadvantage of this strategy is that it may take time to see growth.

There is also a large opportunity cost to invest in a stock long-term instead of investing in a stock short-term may give higher, quicker, returns.

1.3 INTRODUCTION: CONSTANT-MIX (1/2)

Constant-mix is an investment strategy where one buys a ratio of different assets and maintains that ratio.

Maintenance of the ratio is known as rebalancing. Through rebalancing, constant mix allows one to practice the “buy low and sell high” strategy. The asset that performs the best is sold to buy the stock that performs worst.

A common constant-mix that will be used in this study is ratio is 60% stocks and 40% bonds.

There is also many ways to rebalance, but calendar rebalancing will be practiced in this study.

Calendar rebalancing is a simple rebalancing approach that seeks rebalancing at fixed intervals such as quarterly or monthly.

1.3 INTRODUCTION: CONSTANT-MIX (2/2)

Constant-mix success lies in the volatility of the market, by taking advantage of market lows and selling during market highs.

One disadvantage of constant-mix is that the strategy is more involved. It requires constant adjustment to maintain the ratio and maximize gains.

1.4 INTRODUCTION: ASSET ALLOCATION

(1/4)

Each type of asset has a different set of risks and returns, depending on the direction of the market. Therefore, asset allocation is an important part of investing.

Asset allocation is the practice of allocating investments into different asset categories in order to reduce risk.

The assets that will be focused on in this study will be ETFs and bonds.

1.4 INTRODUCTION: ASSET ALLOCATION

(2/4)

ETFs

An exchange-traded fund (ETF) is a type of fund that bundles different types of stock into one fund which one can buy and sell like stock. It is a simple way to participate in asset allocation and reduce risk.

Bonds

Bonds are loans issued by governments and corporations. The loaner gets the return from the interest from the loan and the face value of the loan. Typically, bonds are known to be a less volatile investment than stocks, and give modest returns.

1.4 INTRODUCTION: ASSET ALLOCATION (3/4)

QQQ

Invesco QQQ is an ETF, known for their large investments in Information Technology sector or and technology.

TLT

iShares 20+ Year Treasury Bond ETF (TLT) tracks U.S. Treasury bonds returns. Returns are based on performance of U.S. Treasury market, and is correlated to interest rates.

1.4 INTRODUCTION: ASSET ALLOCATION (4/4)

SPY

The SPDR SP 500 ETF Trust (SPY) is one of the most popular funds and contains 500 large-cap U.S. stocks in the Standard Poor's (SP) 500 Index. It is well-diversified in its sector allocations.

IWN

iShares Russell 2000 Value ETF (IWN) focuses their allocations on small public U.S. companies that might be undervalued by the market, listed on the Russell Index.

2. DATASET DESCRIPTION (1/2)

The dataset used contains has 240 rows and 5 columns.

The 'month' attribute goes from 0 to 239. It represents each month starting from August 2002 to August 2022.

The 'TLT_close' column represents the closing price of TLT at the end of the month, the 'SPY_close' column represents the closing price of SPY at the end of the month, and so on.

The closing price data was sourced from yahoo finance historical stock website. The dataset will be used as reference for calculating returns each month. The dataset can be downloaded [here](#).

2. DATASET DESCRIPTION (2/2)

Dataset View

month	TLT_close	SPY_close	QQQ_close	IWN_close
0	87.080002	91.779999	23.49	39.025002
1	90.400002	81.790001	20.719999	35.775002
2	86.730003	88.519997	24.549999	37.25
3	85.550003	93.980003	27.719999	40.404999
...	...	...	...	...
236	119.449997	412	313.25	184.949997
237	116.559998	412.929993	308.279999	185.309998
238	114.870003	377.25	280.279999	169.360001
239	117.43	411.98999	315.459991	187.25

3. METHODS

1. RETURNS
2. RATE OF RETURN
3. MAX DRAWDOWN
4. BUY-HOLD ALGORITHM
5. CONSTANT-MIX ALGORITHM



3.1 RETURNS

Number of stocks is calculated using contributions (money invested) and the price the stock was bought.

$$stocks = \frac{contributions}{stock_price}$$

The returns is calculated using number of stocks multiplied by the final stock price.

$$returns = stocks * final_price$$

3.2 RATE OF RETURN

Rate of Return (%) or Return on Investment (ROI) is calculated using total contributions and value of portfolio.

$$ROI = \frac{(portfolio_value - total_contributions)}{total_contributions}$$

3.3 MAX DRAWDOWN

Max Drawdown (%) identifies the maximum decline of return on investment. It indicates the largest % loss from the portfolio. It is calculated using the highest return and the lowest return before the investment rises to a peak. This is a common measure to measure how strategies deal with volatility and the largest expected % loss from portfolio or asset.

$$MDD(T) = \frac{P - L}{P}$$

P = peak value

L = lowest value before new peak

3.4 BUY-HOLD ALGORITHM

Input:

price: stock price (of that month)

c: contributions (in this case, $c = \$1,000$)

Output:

total_shares

Method:

For each month,

1. calculate amount of shares, S .
2. add S to *total_shares*.

Calculate the final return based on amount of *total_shares* accumulated over the months.

3.5 CONSTANT-MIX ALGORITHM (1/2)

Input:

s_price : stock price (of that month)
 b_price : bond price (of that month)
 c : contributions (in this case, $c = \$1,000$)

Output:

$total_s_shares$
 $total_b_shares$

Method:

For each month,

1. allocate 40% of the c to bonds and 60% of the c to stock. (in this case, that is \$400 to bonds and \$600 to stocks.)
2. buy bonds shares, B and stock shares, S based on the allocation. Add B to $total_b_shares$ and S to $total_s_shares$.
3. calculate total portfolio value.
4. calculate the % of value in stocks and % of value in bonds inside the portfolio.

3.6 CONSTANT-MIX ALGORITHM (2/2)

1. if % value in stocks is greater than 60%, then sell stock shares, s until % value is back to 60%. Use the money from selling s to buy bond shares b . $total_s_shares$ is deducted s amount and b is added to $total_b_shares$.
2. if % value in stocks is less than 60%, then sell bond shares, b until % value of bond shares is back to 40%. Use the money from selling the bond shares to buy stock shares s . $total_b_shares$ is deducted b amount and $total_s_shares$ increases s amount.
3. if % value in stocks is equals 60%, then no action is needed this month.

Calculate the final return based on amount of $total_s_shares$ and $total_b_shares$ accumulated over the months.

4. RESULTS

1. Constant Mix vs. Buy-Hold strategies: Final Return Summary
 2. Cumulative Returns: Asset Allocation Comparison
 3. Cumulative Returns: Strategy Comparison
 4. Distribution of Rate of Return - Yearly
 5. Asset Allocation Density Plot Comparison
 6. Strategy Density Plot Comparison
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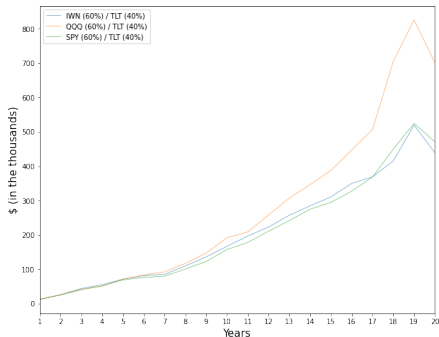
4.1 RESULTS

Constant Mix vs. Buy-Hold strategies Final Return Summary

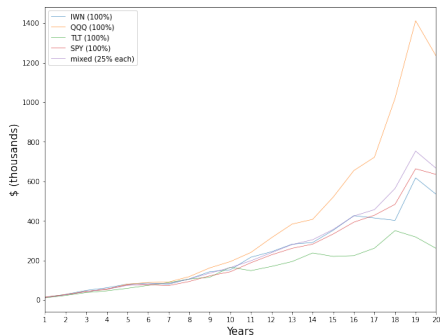
Strategies	Asset Allocation	Final Portfolio Value	Final Rate of Return	Max Drawdown (yearly)
Buy-Hold	IWN (100%)	\$534,367.66	2.23%	-41.68%
	QQQ (100%)	\$1,232,712.10	5.14%	-37.42%
	TLT (100%)	\$260,990.26	1.09%	-22.32%
	SPY (100%)	\$634,919.74	2.65%	-39.33%
	IWN/SPY/QQQ/TLT (25% each)	\$665,747.44	2.77%	-25.32%
Constant-Mix (60/40)	IWN/TLT (60%/40%)	\$439,590.48	1.83%	-22.84%
	QQQ/TLT (60%/40%)	\$701,383.22	2.92%	-26.13%
	SPY/TLT (60%/40%)	\$471,430.20	1.96%	-20.22%

4.2 RESULTS

Cumulative Returns Asset Allocation Comparison



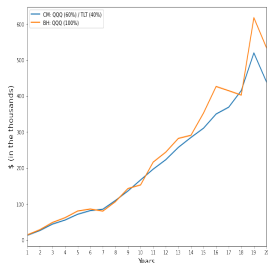
(a) Constant-Mix



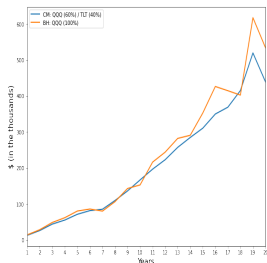
(b) Buy-Hold

4.3 RESULTS

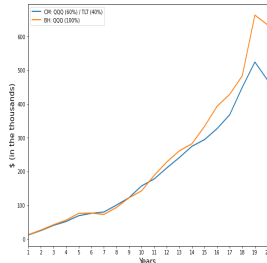
Cumulative Returns Strategy Comparison



(a) QQQ



(b) IWN



(c) SPY

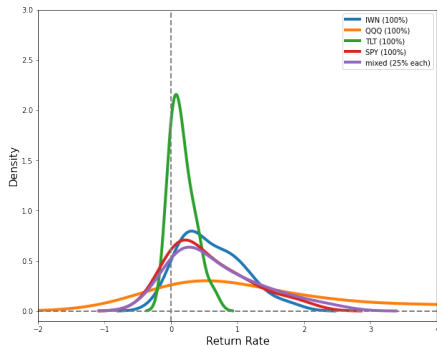
4.4 RESULTS

Distribution of Rate of Return - Yearly Summary Table

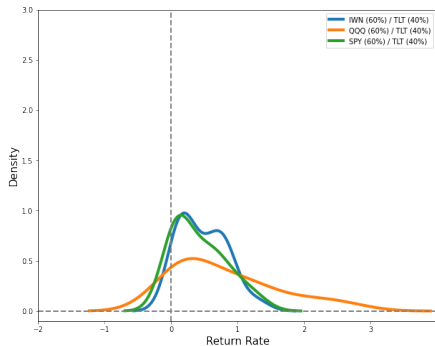
Strategies	Asset Allocation	Mean	Median	Max	Min	Standard Deviation	Skewness	Kurtosis
Buy Hold	IWN (100%)	0.61	0.50	1.71	-0.04	0.46	0.74	0.01
	QQQ (100%)	1.38	0.72	5.19	0.08	1.51	1.33	0.95
	TLT (100%)	0.17	0.12	0.62	-0.06	0.18	0.96	0.54
	SPY (100%)	0.56	0.35	1.91	-0.13	0.57	0.99	0.17
	IWN/SPY/QQQ/TLT (25% each)	0.68	0.44	2.30	-0.01	0.66	1.09	0.43
Constant Mix (60/40)	IWN/TLT (60%/40%)	0.48	0.44	1.28	0.02	0.35	0.54	-0.55
	QQQ/TLT (60%/40%)	0.81	0.59	2.62	0.05	0.78	1.02	0.15
	SPY/TLT (60%/40%)	0.42	0.33	1.30	-0.04	0.40	0.75	-0.43

4.5 RESULTS

Asset Allocation Density Plot Comparison



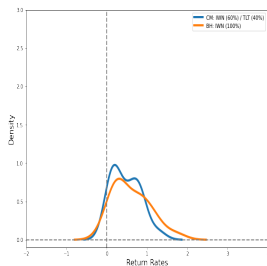
(a) Constant-Mix



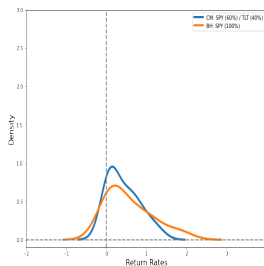
(b) Buy-Hold

4.6 RESULTS

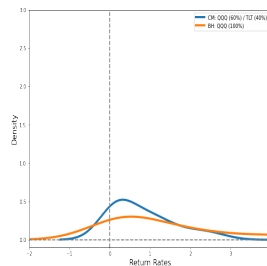
Strategy Density Plot Comparison



(a) QQQ



(b) IWN



(c) SPY

5. CONCLUSION (1/4)

Out of all the stocks, QQQ has the greatest returns over time, but the slightly higher risk than the rest of the stocks.

The stock with the lowest risk is TLT.

Constant-mix with SPY 60% and TLT had the lowest max drawdown or risk out of all of the strategies and asset allocations.

5. CONCLUSION (2/4)

The distributions confirm previous studies results that the constant-mix strategy will mostly likely not yield extreme rates of return.

Constant-mix strategy will result lower extreme gains but also lower extreme losses.

With assets that have higher standard deviation or more volatility such as QQQ, constant-mix can reduce the volatility of those assets but getting high returns from the asset maybe sacrificed.

5. CONCLUSION (3/4)

When comparing constant-mix and buy-hold max drawdown values, constant-mix strategy has significantly lower max drawdown values than buy-hold.

Buy-hold can outperform constant-mix when the asset is high-performing one, such as QQQ.

Constant-mix strategy may hold back the returns of high performing assets, because of the investment of 40% to TLT.

5. CONCLUSION (4/4)

Limitations

The returns from dividends were left out for the sake of simplicity.

Factoring in dividends would greatly increase the returns of each stock and the final return value.

Exact average of annual return can not be pinpointed due to the limited period of time.

6. REFERENCES

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IWN historical closing price. (2022). Yahoo Finance. website.

QQQ historical closing price. (2022). Yahoo Finance. website.

TLT historical closing price. (2022). Yahoo Finance. website.

SPY historical closing price. (2022). Yahoo Finance. website.